B7-H3 (CD276) in Lung Cancer: Reproduction vs. Paper Workflow

Round 2 (spatial + LIANA + full review cycle) - DeepSeek-routed Virtual Biotech, zero changes to source

PAPER'S WORKFLOW (published methods, one-shot)

Fig 4A - Genetics

GWAS credible sets + L2G (germline support)

Fig 4B/4C - scRNA

CELLxGENE Census atlas, UMAP + pseudobulk DE

Fig 4D/S5A - LIANA

CAF-tumor ligand-receptor, high- vs low-CD276

Fig 4E/S5B - Visium

cell2location deconvolution, immune-niche map

Fig 4F - TCGA survival

Cox PH, CD276 quartiles

Tahoe-100M / DepMap (supplemental)

perturbation + essentiality

-> Published figure panels

single synthesized narrative; no visible

adversarial self-check before publication

OUR REPRODUCTION (agentic, Virtual Biotech CSO)

CSO

reads the same source panels/claims,

plans waves of specialist dispatch

single-cell-analyst

scRNA + LIANA + Visium/cell2location

(4B/4C, 4D/S5A, 4E/S5B)

genomics-analyst

GWAS + L2G + ENCODE-rE2G (4A)

clinical-trialist

TCGA survival (4F)

functional-genomics-analyst

Tahoe-100M + DepMap

scientific-reviewer

checks DATA / SCRIPTS / FIGURES directly,

not report prose

Revise -> re-review

4 specialists file evidence-checked fixes ->

ACCEPT WITH DISCLOSED LIMITATIONS

CSO synthesis + claims-evidence ledger

15/15 claims verified

KEY DIFFERENCE: the paper's workflow (as published) is a fixed, one-pass pipeline. Our reproduction adds a built-in adversarial loop (review -> revise -> re-review) that caught a real bug and a real reporting omission BEFORE the result was finalized - see the two callouts below, drawn directly from this run's own logs.

ENGINEERING: a real bug, independently verified

cell2location's first mapping converged correctly, but a pandas column-selection-vs-rename bug made the exported abundance table 100% NaN in all 14 cell-type columns. No checkpoint existed, so the fix required a full retrain. Verified directly against the parquet both times: 100%->0% NaN.

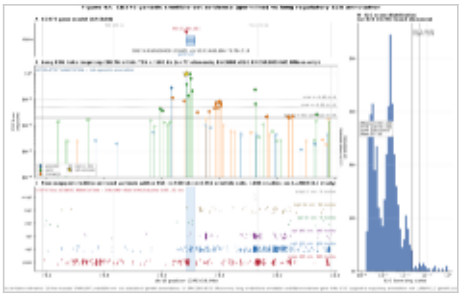
SCIENCE: a real revise-and-resubmit

The first full review did not rubber-stamp the spatial result: it required surfacing donor-concordant immune-compartment depletion (endothelial, NK, macrophage) that was correctly computed but omitted from the narrative in favour of a less robust T-cell/dendritic-cell increase. Fixed before synthesis.

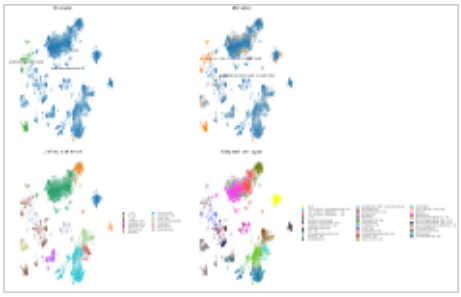
5 run directories - 4 transient network/API interruptions - 0 source-code modifications to TheVirtualBiotech

Key Results: Our Reproduction vs. the Paper's Claims

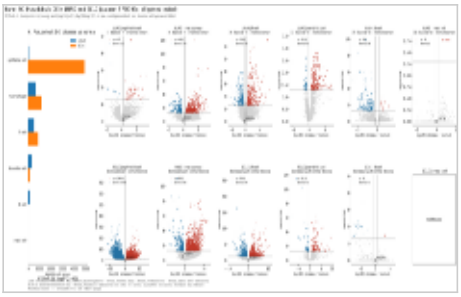
All figures below are this run's own output, unmodified



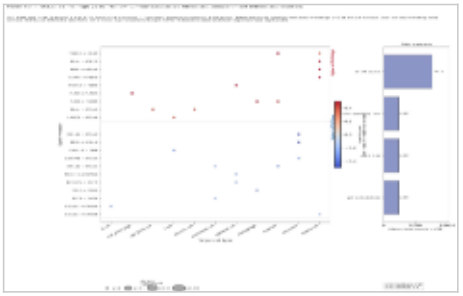
4A - Genetics / E2G regulatory



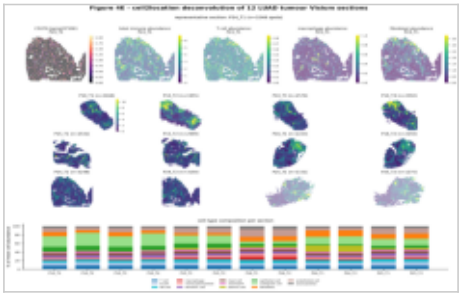
4B - scRNA atlas UMAP



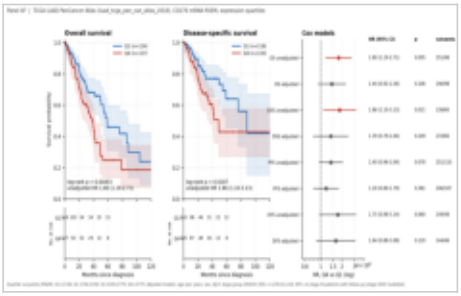
4C - Pseudobulk DE (CD276 not sig.)



4D - LIANA, SCLC CAF B7-H3-hi/lo



4E - cell2location, tumour sections



4F - TCGA LUAD survival

Panel-by-panel: paper's claim vs. this round's finding

Panel	Paper claims	This round found	Verdict
4A	Genetic + regulatory support for CD276	0/8 GWAS, 0/67 L2G; regulatory annotation only (839 linked elements)	Disagree (genetic) Agree (regulatory)
4B/4C	scRNA atlas identifies CD276-associated populations	Atlas reproduced (414K cells); CD276 not sig. in 15/15 DE contrasts (padj>0.29)	Weaker / Disagree
4D/S5A	Signalling unique to B7-H3+ CAFs	Rank-based ECM programme; 55-69% depth gap; no formal test	Weaker / Disagree on "unique"
4E/S5B	Immune-hot niche around high-CD276 spots	Absolute increase is a depth/edge artifact; composition = malignant+, endothelial/NK-	Weaker / Disagree
4F	OS HR 1.62 (p=0.028), DSS HR 1.82 (p=0.031)	Unadj. OS HR 1.80 (p=0.005) reproduces; adjusted OS p=0.106 (n.s.)	Agree unadj. Weaker adj.
Tahoe / DepMap	(target-assessment context)	Tahoe blocked, not assessable; DepMap not essential (CERES -0.230)	Inconclusive Agree

Bottom line: direction mostly agrees with the paper; strength is materially weaker across 4/6 panel groups. CD276 is not tumour-restricted, not differentially expressed, and not independently prognostic once adjusted for stage.

Engineering Note: DeepSeek vs. Claude as the Agent Driver

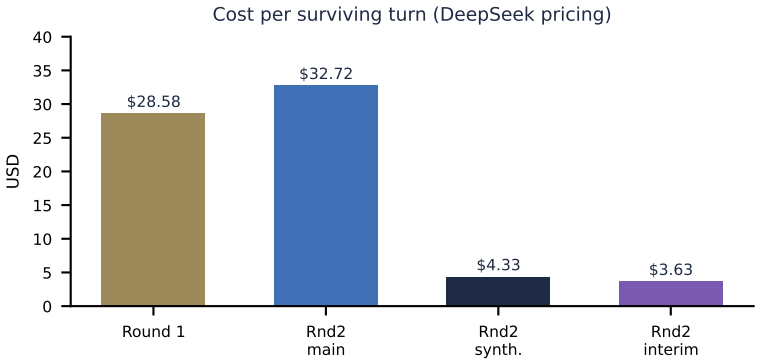
Both rounds ran the unmodified TheVirtualBiotech codebase (claude_agent_sdk) with the model layer routed to DeepSeek's Anthropic-compatible endpoint -- no apples-to-apples Claude run exists in this project.

What worked

Full agentic tool loop (Bash, Read, Write, Edit, Task dispatch, MCP calls) ran through DeepSeek with zero SDK/prompt changes: the CSO pattern is not Claude-specific.

Specialists showed Claude-oriented agentic behaviour: self-corrected a subtle bug, ran a genuine review-revise-reaccept loop, wrote self-critical caveats.

Cost is low: round 2's 3 surviving turns totalled ~$41 (chart, right); round 1's single turn was $28.58.



What's unresolved / caveated

response.model always echoes the REQUESTED id, never a DeepSeek-native name -- it alone cannot verify what actually executed underneath.

Per DeepSeek's documented routing (table, right), the specialist and reviewer tiers land in the same bucket -- the paper's two-tier capability split is not preserved.

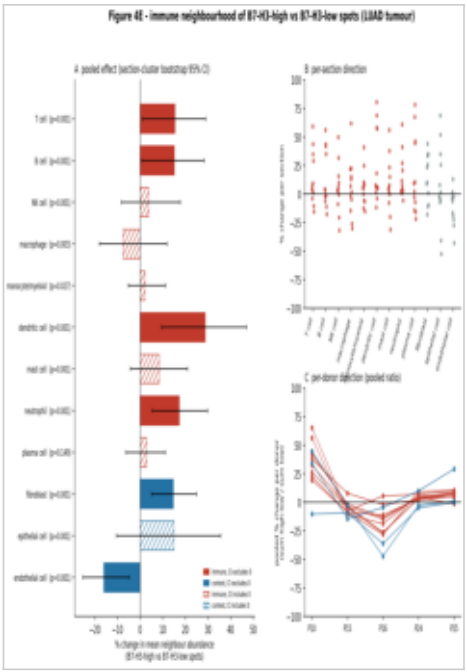
5 run directories, 4 transient interruptions in round 2 -- diagnosed as network/API-level, not DeepSeek-specific (no Claude-routed control run exists to compare against).

Role -> requested model id -> DeepSeek's documented routing

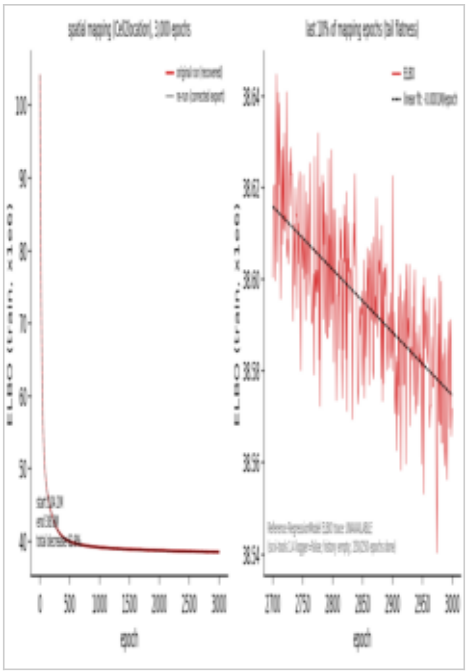
CSO / chief-of-staff	`claude-haiku-4-5`	deepseek-flash bucket
Specialists (8 roles)	`claude-sonnet-4-5`	deepseek-flash bucket (same)
scientific-reviewer	`claude-haiku-4-5`	deepseek-flash bucket

Both tiers converge server-side -- see engineering_validation/test_F_model_audit.py for the full audit trail.

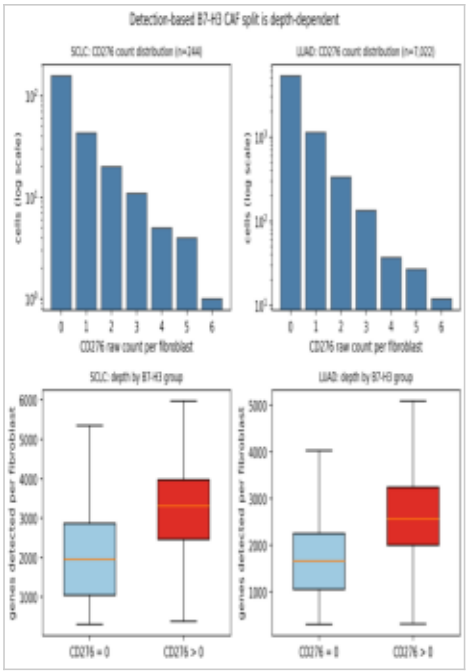
Beyond the paper: extra diagnostics the paper doesn't show



Spatial-neighbourhood immune composition change (not in paper)



cell2location training ELBO convergence diagnostic



CAF B7-H3 detection-depth confound audit

Overall: DeepSeek, swapped in as a drop-in Claude replacement with zero code changes, successfully drove the full multi-agent CSO/specialist/reviewer loop end-to-end -- but the paper's two-tier model-capability split collapses server-side, and reliability could not be benchmarked against a native-Claude control run.